Lecture 12

Vectors, length and projection

Uses.

Summary 1: Ordered pairs have equal corresponding entries. Cartesian products collect such pairs.
Summary 2: Real arithmetic, nonnegative square roots and finite sums are available. Standing fact E gives cosine as a bijection from $[0, \pi]$ onto $[-1, 1]$, with inverse arccos.

Objectives. *A displacement has components, but its length does not depend on the signs of those components. The dot product relates components to lengths and angles. We prove the inequalities that make these definitions consistent, then split a vector into a part along a given direction and an orthogonal remainder. The same split gives the closest point on a line through the origin.*

By the end of this lecture you will be able to:

- Compute vector sums, scalar multiples and standard coordinates in $\mathbb{R}^n$.
- Prove Cauchy–Schwarz and the vector triangle inequality, including their equality cases.
- Compute lengths, orthogonality and angles between nonzero vectors.
- Prove and apply orthogonal projection onto a nonzero direction and its closest-point property.

12.1 Coordinates and displacements

The Cartesian product $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ is the set of ordered pairs (x_1, x_2) of real numbers. The set $\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ consists of ordered triples. More generally, for an integer $n \geq 1$, the set $\mathbb{R}^n$ consists of ordered lists $(x_1, \ldots, x_n)$ with every $x_j \in \mathbb{R}$. Two lists are equal exactly when their corresponding entries are equal. An element of $\mathbb{R}^n$ is a *vector*. Its entries are its *components*. We write geometric vectors in bold, as $\boldsymbol{u} = (u_1, \ldots, u_n)$.

In the coordinate plane and in three-dimensional space, we choose an origin and mutually perpendicular coordinate axes with the same unit of length. A point is specified by its coordinates. A *directed segment* from A to B has starting point A and end point B. Its *displacement* is the vector

$$\overrightarrow{AB} = (b_1 - a_1, \ldots, b_n - a_n).$$

Here A has coordinates $(a_1, \ldots, a_n)$ and B has coordinates $(b_1, \ldots, b_n)$, with $n = 2$ or $n = 3$. Segments with the same displacement represent the same vector even when their starting points differ. The *position vector* of a point is its displacement from the origin.

For example, take points with coordinates

$$A = (1, 2, -1), \qquad B = (4, -2, 5).$$

Their displacement is $(4 - 1, -2 - 2, 5 - (-1)) = (3, -4, 6)$. Reversing the endpoints gives $(-3, 4, -6)$. A point and a displacement can have the same coordinate list, but they play different roles: a point specifies a location, while a displacement specifies a change.

12.2 Vector operations and standard coordinates

Let $\boldsymbol{u}$ and $\boldsymbol{v}$ belong to the same $\mathbb{R}^n$, and let $\lambda \in \mathbb{R}$. Their *sum* and the *scalar multiple* of $\boldsymbol{u}$ are defined by

$$\boldsymbol{u} + \boldsymbol{v} = (u_1 + v_1, \ldots, u_n + v_n), \qquad \lambda \boldsymbol{u} = (\lambda u_1, \ldots, \lambda u_n).$$

A *scalar* here means a real number. The *zero vector* is $\mathbf{0} = (0, \ldots, 0)$. The opposite vector is $-\boldsymbol{u} = (-1)\boldsymbol{u}$, and subtraction means $\boldsymbol{u} - \boldsymbol{v} = \boldsymbol{u} + (-\boldsymbol{v})$. Only vectors with the same number of components are added.

These operations inherit the arithmetic laws component by component. For instance, the jth component of $(\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}$ is $(u_j + v_j) + w_j = u_j + (v_j + w_j)$, so addition is associative. The equality $u_j + v_j = v_j + u_j$ proves commutativity. The identities $\lambda(u_j + v_j) = \lambda u_j + \lambda v_j$ and $(\lambda + \mu)u_j = \lambda u_j + \mu u_j$ prove the two distributive laws. Similarly, $\lambda(\mu u_j) = (\lambda\mu)u_j$ proves compatibility of successive scalar multiplications. Equality of all components gives each vector identity.

Geometrically, adding displacements joins two successive directed segments. Indeed, for points A, B and C, component subtraction gives

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}.$$

A positive scalar stretches a segment without reversing it. A negative scalar reverses its direction as well as changing its length. Multiplication by zero gives the zero displacement.

A *linear combination* of vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ is a sum $c_1\boldsymbol{v}_1 + \cdots + c_k\boldsymbol{v}_k$ with coefficients $c_j \in \mathbb{R}$. The *standard coordinate vector* $\boldsymbol{e}_j \in \mathbb{R}^n$ has entry 1 in position j and entry 0 elsewhere. Every vector has the representation

$$\boldsymbol{u} = u_1\boldsymbol{e}_1 + \cdots + u_n\boldsymbol{e}_n. \tag{12.1}$$

Its jth component is u_j, which proves the equality. This representation is unique: the jth component of a combination $c_1\boldsymbol{e}_1 + \cdots + c_n\boldsymbol{e}_n$ is c_j, so equality with $\boldsymbol{u}$ forces $c_j = u_j$ for every j. The coefficients are the *coordinates in the standard basis*. Here this phrase refers precisely to the unique representation (12.1). Lecture 18 gives the general definition of a basis.

Example 12.1 (A fraction of a displacement)**.** For the points A and B above, find the point a quarter of the way from A to B. Its position is obtained by adding a quarter of the displacement to the position of A:

$$(1, 2, -1) + \frac{1}{4}(3, -4, 6) = \left(\frac{7}{4}, 1, \frac{1}{2}\right).$$

For a general parameter $t \in \mathbb{R}$, the position $A + t(B - A)$ equals A at $t = 0$ and B at $t = 1$. Parameters $0 \le t \le 1$ describe the segment between these points. Parameters outside that interval continue the same straight line past an endpoint.

12.3 The dot product and length

Definition 12.2 (Dot product and Euclidean norm)**.** For $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$, their *dot product* is the real number

$$\boldsymbol{u} \cdot \boldsymbol{v} = \sum_{j=1}^{n} u_j v_j.$$

The *Euclidean norm*, or length, of $\boldsymbol{u}$ is

$$\|\boldsymbol{u}\| = \sqrt{\boldsymbol{u} \cdot \boldsymbol{u}} = \sqrt{u_1^2 + \cdots + u_n^2}.$$

The distance between points with position vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ is $\|\boldsymbol{b} - \boldsymbol{a}\|$.

The dot product is symmetric, since $u_j v_j = v_j u_j$. It is linear in each argument. For example, distributing each product in the finite sum gives

$$(\lambda \boldsymbol{u} + \mu \boldsymbol{v}) \cdot \boldsymbol{w} = \lambda (\boldsymbol{u} \cdot \boldsymbol{w}) + \mu (\boldsymbol{v} \cdot \boldsymbol{w}).$$

Symmetry gives the same rule for the second argument. We call this pair of linearity properties *bilinearity*. The expression $\boldsymbol{u} \cdot \boldsymbol{u}$ is a sum of nonnegative squares. It is zero exactly when every component is zero. Consequently,

$$\|\boldsymbol{u}\| \geq 0, \qquad \|\boldsymbol{u}\| = 0 \iff \boldsymbol{u} = \boldsymbol{0}, \qquad \|\lambda \boldsymbol{u}\| = |\lambda| \, \|\boldsymbol{u}\| .$$

The last identity follows by taking the nonnegative square root of $\lambda^2 (\boldsymbol{u} \cdot \boldsymbol{u})$.

Example 12.3 (Length and a dot product)**.** Take the vectors

$$\boldsymbol{u} = (3, 4, 12), \qquad \boldsymbol{v} = (2, -1, 2).$$

Their squared lengths and dot product are

$$\|\boldsymbol{u}\|^2 = 9 + 16 + 144 = 169, \qquad \|\boldsymbol{v}\|^2 = 4 + 1 + 4 = 9, \qquad \boldsymbol{u} \cdot \boldsymbol{v} = 6 - 4 + 24 = 26.$$

Thus their lengths are 13 and 3. The dot product is a scalar, not another vector. Its value will determine the angle and the projection below.

12.4 Length inequalities and angles

A finite family of vectors is *linearly dependent* if some real coefficients, not all zero, give a zero linear combination. It is *linearly independent* if a zero linear combination forces every coefficient to be zero. Thus independence is the negation of dependence. For a pair of vectors, dependence means that one is a scalar multiple of the other, including pairs containing zero. Indeed, if $a\boldsymbol{u} + b\boldsymbol{v} = \boldsymbol{0}$ and $a \neq 0$, then $\boldsymbol{u} = -(b/a)\boldsymbol{v}$. If $a = 0$, then $b \neq 0$ and $\boldsymbol{v} = \boldsymbol{0}$. The converse follows by moving a scalar multiple to one side.

Theorem 12.4 (Cauchy–Schwarz inequality). *For $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$,*

$$|\boldsymbol{u} \cdot \boldsymbol{v}| \leq \|\boldsymbol{u}\| \, \|\boldsymbol{v}\| .$$

Equality holds exactly when the two vectors are linearly dependent.

Proof. If $\boldsymbol{v} = \boldsymbol{0}$, both sides are zero and the pair is dependent. Suppose $\boldsymbol{v} \neq \boldsymbol{0}$. Choose the scalar $\alpha = (\boldsymbol{u} \cdot \boldsymbol{v})/\|\boldsymbol{v}\|^2$. Bilinearity gives

$$0 \leq \|\boldsymbol{u} - \alpha \boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 - 2\alpha(\boldsymbol{u} \cdot \boldsymbol{v}) + \alpha^2 \|\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 - \frac{(\boldsymbol{u} \cdot \boldsymbol{v})^2}{\|\boldsymbol{v}\|^2}.$$

Multiplying by the positive squared norm proves the squared inequality. Taking nonnegative square roots gives the stated inequality. Equality holds exactly when $\boldsymbol{u} - \alpha \boldsymbol{v} = \boldsymbol{0}$. For nonzero $\boldsymbol{v}$ this is exactly dependence of the pair. ∎

For example, take $\boldsymbol{a} = (-2, 1)$ and $\boldsymbol{b} = (4, -2) = -2\boldsymbol{a}$. Their dot product is -10, while their lengths are $\sqrt{5}$ and $2\sqrt{5}$. Cauchy–Schwarz is an equality after taking the absolute value. A negative dot product does not prevent that equality.

For nonzero vectors, Cauchy–Schwarz places $(\boldsymbol{u} \cdot \boldsymbol{v})/(\|\boldsymbol{u}\| \, \|\boldsymbol{v}\|)$ in $[-1, 1]$. By standing fact E, precisely one $\theta \in [0, \pi]$ has that cosine. The *angle between the vectors* is therefore defined by

$$\theta = \arccos \frac{\boldsymbol{u} \cdot \boldsymbol{v}}{\|\boldsymbol{u}\| \, \|\boldsymbol{v}\|}. \tag{12.2}$$

For vectors in the coordinate plane or space, this agrees with the usual geometric angle: expand $\|\boldsymbol{u} - \boldsymbol{v}\|^2$ and compare with the school law of cosines. Its value is measured in radians. An angle with a zero vector is undefined. Two nonzero vectors are *parallel* if one is a scalar multiple of the other. They have the *same direction* if that scalar is positive. They have *opposite directions* if that scalar is negative. The corresponding angles are 0 and π. For the vectors $(3, 4, 12)$ and $(2, -1, 2)$, the angle is $\arccos(26/(13 \cdot 3)) = \arccos(2/3)$.

Theorem 12.5 (Triangle inequality). *For $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$,*

$$\|\boldsymbol{u} + \boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\| .$$

Equality holds exactly when one vector is zero or the two nonzero vectors have the same direction.

Proof. Expanding the square and applying Cauchy–Schwarz gives

$$\|\boldsymbol{u} + \boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + 2\boldsymbol{u} \cdot \boldsymbol{v} + \|\boldsymbol{v}\|^2 \leq \|\boldsymbol{u}\|^2 + 2 \|\boldsymbol{u}\| \, \|\boldsymbol{v}\| + \|\boldsymbol{v}\|^2 .$$

The right side is $(\|\boldsymbol{u}\| + \|\boldsymbol{v}\|)^2$. Taking nonnegative square roots proves the inequality. If either vector is zero, equality holds. For nonzero vectors, equality requires $\boldsymbol{u} \cdot \boldsymbol{v} = \|\boldsymbol{u}\| \, \|\boldsymbol{v}\|$. Cauchy–Schwarz then makes the vectors proportional. Writing $\boldsymbol{u} = \lambda \boldsymbol{v}$ shows that the required equality is $\lambda \|\boldsymbol{v}\|^2 = |\lambda| \, \|\boldsymbol{v}\|^2$. Since $\lambda \neq 0$, this holds exactly when $\lambda > 0$. ∎

For the opposite pair $\boldsymbol{a}$ and $\boldsymbol{b}$ above, the sum is $(2, -1)$. Its length is $\sqrt{5}$, strictly less than the sum $3\sqrt{5}$ of their lengths. Thus the equality conditions for these two inequalities differ. Applied to $\boldsymbol{b} - \boldsymbol{a} = (\boldsymbol{b} - \boldsymbol{c}) + (\boldsymbol{c} - \boldsymbol{a})$, the triangle inequality says that going through a third point cannot shorten the direct distance. The same inequality also bounds changes in length. Apply it to $\boldsymbol{u} = (\boldsymbol{u} - \boldsymbol{v}) + \boldsymbol{v}$ to obtain $\|\boldsymbol{u}\| - \|\boldsymbol{v}\| \leq \|\boldsymbol{u} - \boldsymbol{v}\|$. Interchanging the vectors proves the *reverse triangle inequality*

$$\big|\|\boldsymbol{u}\| - \|\boldsymbol{v}\|\big| \leq \|\boldsymbol{u} - \boldsymbol{v}\| .$$

12.5 Orthogonality and projection

Two vectors are *orthogonal* if their dot product is zero. For nonzero vectors, this means their angle is $\pi/2$. The zero vector is orthogonal to every vector, even though it has no angle. Expanding the square as above shows the Pythagorean identity:

$$\boldsymbol{p} \cdot \boldsymbol{w} = 0 \quad \Longrightarrow \quad \|\boldsymbol{p} + \boldsymbol{w}\|^2 = \|\boldsymbol{p}\|^2 + \|\boldsymbol{w}\|^2 .$$

For a nonzero vector $\boldsymbol{v}$, the set $L = \{t\boldsymbol{v} : t \in \mathbb{R}\}$ is the line through the origin in its direction. The *orthogonal projection* of $\boldsymbol{u}$ onto this line is a vector $\boldsymbol{p} \in L$ such that $\boldsymbol{u} - \boldsymbol{p}$ is orthogonal to $\boldsymbol{v}$. Figure 12.1 shows the two parts of the displacement.

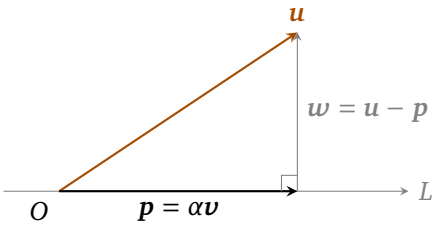


Figure 12.1: The projection lies on the chosen line. The remaining displacement is orthogonal to that line. The picture uses a positive coefficient; a projection can also point in the opposite direction.

Theorem 12.6 (Projection and the closest point)**.** *Let $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ with $\boldsymbol{v} \neq \boldsymbol{0}$. The orthogonal projection onto $L = \{t\boldsymbol{v} : t \in \mathbb{R}\}$ exists and is unique. It is*

$$\boldsymbol{p} = \frac{\boldsymbol{u} \cdot \boldsymbol{v}}{\|\boldsymbol{v}\|^2}\boldsymbol{v}.$$

Among all vectors on L, it is the unique one that minimises the distance to $\boldsymbol{u}$.

Proof. Write a candidate as $\boldsymbol{p} = \alpha\boldsymbol{v}$. Orthogonality requires $(\boldsymbol{u} - \alpha\boldsymbol{v}) \cdot \boldsymbol{v} = \boldsymbol{u} \cdot \boldsymbol{v} - \alpha \|\boldsymbol{v}\|^2 = 0$. The nonzero squared norm gives exactly one coefficient, $\alpha = (\boldsymbol{u} \cdot \boldsymbol{v})/\|\boldsymbol{v}\|^2$. This proves existence and uniqueness.

Set $\boldsymbol{w} = \boldsymbol{u} - \alpha\boldsymbol{v}$. For any $t \in \mathbb{R}$, $\boldsymbol{u} - t\boldsymbol{v} = \boldsymbol{w} + (\alpha - t)\boldsymbol{v}$ is a sum of orthogonal vectors. The Pythagorean identity gives

$$\|\boldsymbol{u} - t\boldsymbol{v}\|^2 = \|\boldsymbol{w}\|^2 + (\alpha - t)^2 \|\boldsymbol{v}\|^2 . \tag{12.3}$$

The second term is nonnegative and vanishes exactly at $t = \alpha$. Taking square roots preserves the minimum and its uniqueness. ∎

Example 12.7 (A projection and its error)**.** Find the projection of $\boldsymbol{u} = (3, 4, 12)$ onto the line in direction $\boldsymbol{v} = (2, -1, 2)$. Their dot product is 26 and $\|\boldsymbol{v}\|^2 = 9$. The coefficient and projection are

$$\alpha = \frac{26}{9}, \qquad \boldsymbol{p} = \left(\frac{52}{9}, -\frac{26}{9}, \frac{52}{9}\right).$$

Subtracting gives the orthogonal remainder

$$\boldsymbol{w} = \boldsymbol{u} - \boldsymbol{p} = \left(-\frac{25}{9}, \frac{62}{9}, \frac{56}{9}\right).$$

Its dot product with $\boldsymbol{v}$ is $(-50-62+112)/9=0$, as required. Its squared norm is $(625+3844+3136)/81=845/9$. Thus the distance to the line is $13\sqrt{5}/3$. The squared projection length is $\alpha^2\|\boldsymbol{v}\|^2=676/9$, and $676/9+845/9=169=\|\boldsymbol{u}\|^2$ verifies the decomposition. Every other coefficient adds the positive error term in (12.3).

A *unit vector* is a vector of norm 1. The vector $\widehat{\boldsymbol{v}}=\boldsymbol{v}/\|\boldsymbol{v}\|$ is the unit vector in the direction of nonzero $\boldsymbol{v}$. The *signed scalar component* of $\boldsymbol{u}$ in that direction is

$$s=\boldsymbol{u}\cdot\widehat{\boldsymbol{v}}=\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{\|\boldsymbol{v}\|},\qquad \boldsymbol{p}=s\widehat{\boldsymbol{v}},\qquad \|\boldsymbol{p}\|=|s|.$$

For the worked pair, this component is $26/3$. Its sign records direction, whereas the projection length is nonnegative. For example, projecting $(-3,4)$ onto the direction $(1,0)$ gives scalar component -3, vector $(-3,0)$ and length 3. These quantities will enter the definition of constant-force work in Lecture 13.

12.6 Coordinates along perpendicular axes

An *orthonormal pair* consists of two orthogonal unit vectors. In the plane, take real numbers a and b with $a^2+b^2=1$ and set

$$\boldsymbol{q}_1=(a,b),\qquad \boldsymbol{q}_2=(-b,a).$$

Their squared norms are both $a^2+b^2=1$, and their dot product is $-ab+ba=0$. Thus they form an orthonormal pair. For any $\boldsymbol{u}=(x,y)$, its signed components along these two directions are

$$c_1=\boldsymbol{u}\cdot\boldsymbol{q}_1=ax+by,\qquad c_2=\boldsymbol{u}\cdot\boldsymbol{q}_2=-bx+ay.$$

The corresponding vector sum is exactly $\boldsymbol{u}$, because

$$c_1\boldsymbol{q}_1+c_2\boldsymbol{q}_2=\big((a^2+b^2)x,(a^2+b^2)y\big)=(x,y).$$

Taking dot products with $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$ shows that no other coefficients can give the same vector. The Pythagorean identity gives $\|\boldsymbol{u}\|^2=c_1^2+c_2^2$. Changing these perpendicular unit axes changes the coordinates but preserves the length. The perpendicularity and unit-length hypotheses are both used.

For example, let $\boldsymbol{q}_1=(3/5,4/5)$ and $\boldsymbol{q}_2=(-4/5,3/5)$. For the vector $\boldsymbol{u}=(2,1)$, the signed components are

$$c_1=\frac{6}{5}+\frac{4}{5}=2,\qquad c_2=-\frac{8}{5}+\frac{3}{5}=-1.$$

Hence $\boldsymbol{u}=2\boldsymbol{q}_1-\boldsymbol{q}_2$. Its squared length is $2^2+(-1)^2=5$, which agrees with its original coordinates. The numbers 2 and -1 are coordinates in the new axes, not entries in the original coordinate list.

Summary 12.

- Vectors in $\mathbb{R}^n$ are ordered real lists. Addition and scalar multiplication act component by component. The standard coordinate representation (12.1) is unique.
- Definition 12.2: the dot product is symmetric and bilinear. The Euclidean norm is nonnegative, vanishes only at zero and obeys $\|\lambda \boldsymbol{u}\| = |\lambda| \, \|\boldsymbol{u}\|$.
- Theorems 12.4 and 12.5: Cauchy–Schwarz has equality for dependent pairs. The triangle inequality has equality only for a zero vector or nonzero vectors in the same direction.
- Equation (12.2) defines an angle only for two nonzero vectors. Orthogonality means zero dot product and includes the zero vector.
- Theorem 12.6: projection onto a nonzero direction gives an orthogonal decomposition and the unique closest point on its line. The signed scalar component differs from its absolute length.

Sheet 12 develops component calculations, distance minimisation and a given model for the centre of mass of point particles. Lecture 13 uses dot products for work and introduces oriented area and torque.

References

J. Stewart, *Calculus: Early Transcendentals*, 8th ed., Cengage, 2016, §§12.1–12.3. K. F. Riley, M. P. Hobson and S. J. Bence, *Mathematical Methods for Physics and Engineering*, 3rd ed., Cambridge University Press, 2006, §7.6.